MALL CUSTOMER CLUSTERING USING MACHINE LEARNING

PROJECT REPORT

A Unsupervised Learning Approach for Customer Segmentation & Target Marketing

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CERTIFICATE OF ORIGINALITY

This is to certify that the major project report entitled "Mall Customer Clustering Using Machine Learning: An Unsupervised Learning Approach for Customer Segmentation & Target Marketing" submitted by Person bearing Registration Number Person in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science & Engineering from Chandigarh University, Gharuan, is an authentic record of original research carried out under my supervision and guidance.

To the best of my knowledge, the matter embodied in this report has not been submitted to any other University or Institute for the award of any degree, diploma, or fellowship.

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ABSTRACT

In competitive modern retail environments, modern commercial centers must optimize operational allocation and personalized customer outreach. Customer segmentation divides a heterogeneous customer base into smaller, homogeneous clusters based on shared purchase behaviors, demographic parameters, and engagement metrics.

This project delivers an end-to-end unsupervised machine learning pipeline applied to commercial mall customer data. Using the Mall Customer dataset, the study applies exploratory data analysis (EDA), statistical preprocessing, and dimensionality evaluation. Two primary clustering techniques are implemented and compared: K-Means Clustering and Hierarchical Agglomerative Clustering. Optimal cluster selection ($k=5$) is verified using the Within-Cluster Sum of Squares (WCSS via the Elbow Method) and Silhouette Coefficient validation ($s \approx 0.554$). Principal Component Analysis (PCA) validates spatial separation across lower dimensions.

The study categorizes the customer base into five distinct profiles: Targeted VIPs (High Income, High Spending), Targeted Savers (High Income, Low Spending), Impulse Shoppers (Low Income, High Spending), Budget Shoppers (Low Income, Low Spending), and General Shoppers (Moderate Income, Moderate Spending). Finally, the report outlines concrete retail operations strategies, marketing interventions, and future opportunities to integrate deep autoencoders and temporal transaction logs.

TABLE OF CONTENTS

- Introduction

  - 1.1 Background and Context
  - 1.2 Problem Statement
  - 1.3 Project Objectives
  - 1.4 Scope of the Study

- Literature Review

  - 2.1 Unsupervised Clustering in Retail Operations
  - 2.2 Comparative Review of Clustering Methodologies

- Dataset Profile & Exploratory Data Analysis

  - 3.1 Dataset Schema & Attribute Descriptions
  - 3.2 Preprocessing, Cleaning, and Scaling
  - 3.3 Statistical Distribution Analysis

- Methodology & Algorithm Implementation

  - 4.1 K-Means Clustering Mathematical Formulation
  - 4.2 Optimal Cluster Selection (Elbow Method & Silhouette Metric)
  - 4.3 Agglomerative Hierarchical Clustering & Dendrogram Formulation

- Results & Segmentation Analytics

  - 5.1 Overview of Five-Cluster Segmentation Matrix
  - 5.2 Deep-Dive Cluster Profiles

- Strategic Business Recommendations

  - 6.1 Cluster-Specific Engagement Plans
  - 6.2 Retail Mall Floor Placement & Space Optimization

- Conclusion & Future Scope

  - 7.1 Research Summary
  - 7.2 Future Enhancements

- References

1. INTRODUCTION

1.1 Background and Context

Modern retail management relies heavily on consumer data analytics to maximize lifetime value and enhance shopper satisfaction. Traditional blanket promotional campaigns typically yield low conversion rates and inefficient marketing spending because they overlook variance in customer preferences, income levels, and purchasing behaviors. Unsupervised learning helps physical retail managers uncover customer cohorts without relying on historical labels.

1.2 Problem Statement

Retail mall operators often lack labeled behavioral histories, making it difficult to systematically group visitors by purchasing power and store interactions. Without accurate segmentation, retail spaces risk deploying mismatched retail store mixes, misallocating loyalty benefits, and wasting marketing capital on broad-spectrum campaigns.

1.3 Project Objectives

- Construct an end-to-end data processing and unsupervised learning pipeline for retail mall patrons.
- Quantitatively determine optimal cluster partitions using the Elbow Method and Silhouette Analysis.
- Contrast partition-based clustering (K-Means) with hierarchical clustering (Agglomerative).
- Map resulting multidimensional clusters to actionable retail management and marketing strategies.

1.4 Scope of the Study

The empirical analysis uses mall patron attributes: Age, Gender, Annual Income (measured in thousands of USD), and an engineered Spending Score (scaled 1–100, calculated from aggregated spending events, frequency, and store visit intervals).

2. LITERATURE REVIEW

2.1 Unsupervised Clustering in Retail Operations

Customer segmentation has evolved from simple demographic cross-tabulation to multivariate unsupervised machine learning. Clustering models partition data points so intra-cluster similarity is maximized while inter-cluster similarity is minimized.

2.2 Comparative Review of Clustering Methodologies

Clustering Algorithm	Core Mechanism	Computational Complexity	Handling Non-Spherical Shapes	Outlier Sensitivity
K-Means	Centroid-based partitioning via Euclidean distance minimization	$O(n \cdot k \cdot I \cdot d)$	Poor; assumes isotropic, convex clusters	High; sensitive to extreme anomalies
Agglomerative Hierarchical	Bottom-up merging via linkage criteria (Ward, Complete)	$O(n^2 \log n)$ to $O(n^3)$	Moderate; linkage dependent	Moderate; isolates outliers in late stages
DBSCAN	Density reachability based on ε-neighborhood and MinPts	$O(n \log n)$ to $O(n^2)$	High; handles arbitrary geometries	Low; explicitly tags sparse noise points

While density-based algorithms like DBSCAN excel at handling noise and arbitrary geometries, K-Means remains the retail analytics standard due to its clear centroid interpretability, linear scalability, and straightforward mapping to customer profiles.

3. DATASET PROFILE & EXPLORATORY DATA ANALYSIS (EDA)

3.1 Dataset Schema & Attribute Descriptions

The dataset contains records for 200 distinct mall customers:

- CustomerID: Unique integer identifier assigned to each visiting patron.
- Gender: Categorical variable indicating biological sex (Female / Male).
- Age: Continuous numerical variable indicating patron age in years (18 to 70).
- Annual Income (k$): Continuous numerical variable representing annual earnings.
- Spending Score (1-100): Continuous score assigned by mall management based on observed spend velocity, average order value, and visit cadence.

3.2 Preprocessing, Cleaning, and Scaling

1. Data Sanitization: The dataset contained no missing, null, or inconsistent values across the 200 observations. CustomerID was dropped as an uninformative index.
2. Categorical Encoding: Gender was mapped via binary indicator coding (Female = 0, Male = 1) for correlation checks.
3. Feature Scaling: Distance-based algorithms like K-Means are sensitive to variance magnitudes. While Annual Income ranges from $15k to $137k, Spending Score ranges from 1 to 100. Standard z-score standardization ($z = (x - \mu) / \sigma$) was applied for multi-feature modeling. Bivariate models (Annual Income vs. Spending

Score) were preserved in natural units to keep cluster centroids directly interpretable for business presentations.

3.3 Statistical Distribution Analysis

Metric	Age (Years)	Annual Income ($k)	Spending Score (1-100)
Mean	38.85	60.56	50.20
Standard Deviation	13.97	26.26	25.82
Minimum	18.00	15.00	1.00
25th Percentile (Q_1)	28.75	41.50	34.75
Median (Q_2)	36.00	61.50	50.00
75th Percentile (Q_3)	49.00	78.00	73.00
Maximum	70.00	137.00	99.00

- Key Insight 1: Female patrons represent 56% of the dataset, while male patrons account for 44%.
- Key Insight 2: The distribution of Annual Income is approximately normal, with slight right-skewness and an upper tail extending past $130k.
- Key Insight 3: The bivariate scatter plot of Annual Income versus Spending Score reveals five visual clusters, establishing a strong baseline for algorithm verification.

4. METHODOLOGY & MODEL IMPLEMENTATION

4.1 K-Means Clustering Mathematical Formulation

K-Means partitions n observations into k distinct, non-overlapping clusters $S = {S_1, S_2, \dots, S_k}$. The objective minimizes the Within-Cluster Sum of Squares (WCSS), also called objective function J:

$$J = \sum_{i=1}^{k} \sum_{x \in S_i} | x - \mu_i |^2$$

Where μ_i denotes the centroid of cluster S_i:

$$\mu_i = \frac{1}{|S_i|} \sum_{x \in S_i} x$$

The standard Lloyd's iteration algorithm was executed:

1. Centroid Initialization: Selected via K-Means++ to avoid poor local minima by spreading out initial cluster seeds based on squared distance probability distributions.

2. Assignment Step: Each data point x_j is assigned to its nearest centroid: $$S_i^{(t)} = { x_j : | x_j - \mu_i^{(t)} |^2 \le | x_j - \mu_l^{(t)} |^2 \quad \forall l, 1 \le l \le k }$$
3. Update Step: Recalculate centroids based on all assigned points.
4. Convergence: Steps repeat until centroid shifts fall below threshold $\varepsilon = 10^{-4}$ or maximum iterations reach $I = 300$.

4.2 Optimal Cluster Selection

The Elbow Method (Inertia Analysis)

Inertia was evaluated across candidate cluster counts $k \in [1, 10]$. The reduction in WCSS shows a clear inflection point ("elbow") at $k = 5$. Increasing k beyond 5 yields diminishing reductions in inertia.

Number of Clusters (k)	Within-Cluster Sum of Squares (WCSS)	Marginal Reduction (Δ WCSS)
1	269,981.3	-
2	181,441.2	88,540.1
3	106,345.8	75,095.4
4	73,679.8	32,666.0
5	44,448.5	29,231.3
6	37,233.1	7,215.4
7	30,259.6	6,973.5

Silhouette Coefficient Validation

The Silhouette Coefficient $s(i)$ assesses cluster separation by comparing intra-cluster distance $a(i)$ to nearest-cluster distance $b(i)$:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

The mean Silhouette Coefficient peaked at $k = 5$ with an average value of $s = 0.554$, confirming that $k=5$ provides the strongest cluster separation and internal cohesion.

4.3 Agglomerative Hierarchical Clustering

To validate the K-Means partition structure, Hierarchical Agglomerative Clustering was run using Ward's minimum variance criterion:

$$\Delta D(A, B) = \frac{n_A n_B}{n_A + n_B} | \mu_A - \mu_B |^2$$

Dendrogram analysis confirms that cutting the tree at a Euclidean distance threshold of approximately 150 yields the same five core customer groups identified by K-Means, demonstrating structural stability across methods.

5. RESULTS & SEGMENTATION ANALYTICS

5.1 Overview of Five-Cluster Segmentation Matrix

Cluster ID	Cluster Identifier	Size (N)	Proportion (%)	Mean Age	Mean Annual Income ($k)	Mean Spending Score
Cluster 1	Targeted Savers (Sensible)	35	17.5%	41.11	$88.29	17.11
Cluster 2	Targeted VIPs (Prime)	39	19.5%	32.69	$86.54	82.13
Cluster 3	Impulse Shoppers (Careless)	22	11.0%	25.27	$25.73	79.36
Cluster 4	Budget Shoppers (Cautious)	23	11.5%	45.22	$26.30	20.91
Cluster 5	General Shoppers (Standard)	81	40.5%	40.17	$55.30	49.52

5.2 Deep-Dive Cluster Profiles

Cluster 1: High Income, Low Spending ("Targeted Savers / Sensible")

- Profile: Customers with high disposable income ($88.3k mean) who spend conservatively (score 17.1). The average age is approximately 41 years.
- Behavioral Traits: Research-driven, selective, price-to-value conscious. They tend to bypass promotional hype and impulse displays.

Cluster 2: High Income, High Spending ("Targeted VIPs / Prime Customers")

- Profile: Customers with high annual earnings ($86.5k mean) and strong spending velocity (score 82.1). The average age is approximately 33 years.
- Behavioral Traits: Brand-conscious, trend-driven, and regular consumers of premium retail, experiential entertainment, and fine dining. This segment generates the highest net margin per square foot.

Cluster 3: Low Income, High Spending ("Impulse Buyers / Careless")

- Profile: Customers with modest income ($25.7k mean) but high spending habits (score 79.4). This is the youngest group, averaging 25.3 years.
- Behavioral Traits: Highly influenced by youth fashion trends, promotional deals, pop-up events, and digital storefront marketing.

Cluster 4: Low Income, Low Spending ("Budget Buyers / Cautious")

- Profile: Older demographic (45.2 years mean) with lower income ($26.3k mean) and low spend indices (score 20.9).
- Behavioral Traits: Frugal, utility-driven visits focused on essential groceries and discount staples.

Cluster 5: Average Income, Average Spending ("General Shoppers / Standard")

- Profile: The largest cohort (40.5% of total visitors), with median income ($55.3k) and median spending (score 49.5).
- Behavioral Traits: Reliable, predictable family footfall. Responsive to seasonal sales, back-to-school periods, and family bundle discounts.

6. STRATEGIC BUSINESS RECOMMENDATIONS

6.1 Cluster-Specific Engagement Plans

- For Cluster 2 (VIPs / Prime):
    - Introduce an exclusive, invite-only VIP Concierge program with personal styling services, reserved priority parking, and private product showcases.
    - Curate luxury pop-ups, premium dining partnerships, and invitations to early product releases.
- For Cluster 1 (Targeted Savers):
    - Shift messaging from brand prestige to functional utility, product durability, and long-term value.
    - Provide transparent product reviews, premium warranty extensions, and tax-advantaged financing options on high-ticket luxury investments.
- For Cluster 3 (Impulse Shoppers):
    - Feature accessible fast-fashion options, digital payment integrations, buy-now-pay-later (BNPL) checkout options, and interactive social-media photo installations.
    - Implement limited-time flash discounts and gamified mobile app notifications to encourage spontaneous store visits.
- For Cluster 4 (Budget Shoppers):
    - Emphasize value-first retail formats: dollar stores, bulk grocers, discount outlets, and price-match guarantees.
    - Distribute print and digital discount circulars focused on essential household staples.
- For Cluster 5 (General Shoppers):

- Implement points-based loyalty programs that reward sustained, incremental spending across multi-category tenants (apparel, groceries, and entertainment).
- Run seasonal mall-wide family events, holiday festivals, and weekend dining promotions.

6.2 Retail Mall Floor Placement & Space Optimization

- Zone Anchor Tenants: Position luxury brands (appealing to Clusters 1 and 2) along main mezzanine levels, alongside fine dining to encourage dwell time.
- Fast-Casual & Trend Corridors: Place trendy youth fashion, entertainment arcades, and affordable snack kiosks (targeting Cluster 3) near main entranceways and transit exits.
- Community Hubs: Anchor lower levels with large-format grocery stores and value retailers to serve Clusters 4 and 5 efficiently.

7. CONCLUSION & FUTURE SCOPE

7.1 Research Summary

This project built and validated an unsupervised machine learning pipeline for retail customer segmentation. By applying K-Means clustering alongside Hierarchical Agglomerative Clustering to the Mall Customer dataset, we identified five distinct customer segments ($k = 5$), confirmed by the Elbow Method and Silhouette Analysis ($s = 0.554$). This segmentation provides mall management with an empirical framework to replace generic promotions with targeted tenant placement and customer retention strategies.

7.2 Future Enhancements

- Temporal Dimension Integration: Augment static demographic cross-sections with longitudinal point-of-sale (POS) and timestamped foot-traffic tracking (RFM: Recency, Frequency, Monetary value).
- Non-Linear Dimensionality Reduction: Evaluate deep learning Autoencoders and Uniform Manifold Approximation and Projection (UMAP) for large-scale, high-dimensional consumer datasets.
- Real-Time Recommendation Engines: Connect cluster classification to mobile geofencing to trigger context-aware promotions as patrons move through different mall zones.

8. REFERENCES

- [1] J. MacQueen, "Some Methods for classification and Analysis of Multivariate Observations," in *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 1, pp. 281–297, 1967.
- [2] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, Nov. 1987.

- [3] D. Arthur and S. Vassilvitskii, "k-means++: The advantages of careful seeding," in *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pp. 1027–1035, 2007.
- [4] A. K. Jain, "Data clustering: 50 years beyond K-means," *Pattern Recognition Letters*, vol. 31, no. 8, pp. 651–666, 2010.
- [5] L. Kaufman and P. J. Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*, John Wiley & Sons, New York, 2009.
- [6] E. W. Ngai, L. Xiu, and D. C. Chau, "Application of data mining techniques in customer relationship management: A literature review and classification," *Expert Systems with Applications*, vol. 36, no. 2, pp. 2592–2602, 2009.
- [7] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.